# Module 3: Probability and Distributions

**Five lessons | From event rules to probability models**

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## Learning sequence

Module 2 described recorded data. This module defines uncertainty about a selection or future outcome, then calculates probabilities under explicit models. Finish the descriptive-statistics review first.

1. **Probability rules:** outcomes, events, complements, intersections, and unions.
2. **Conditional probability:** reference groups, independence, multiplication, and Bayes' theorem.
3. **Discrete random variables:** a probability mass function, cumulative probabilities, expected value, and variance.
4. **Binomial and Poisson models:** assumptions, count probabilities, exposure, and simulation.
5. **Continuous distributions:** uniform areas, normal tails, standardization, and quantiles.

## Lessons and practice

| Lesson | Main question | Practice |
|---|---|---|
| 1. Probability rules | What event and selection mechanism define the probability? | Lesson 1 practice |
| 2. Conditional probability | How does new information change the reference group? | Lesson 2 practice |
| 3. Discrete random variables | What values can occur, with what probabilities? | Lesson 3 practice |
| 4. Binomial and Poisson models | Which count mechanism and event apply? | Lesson 4 practice |
| 5. Continuous distributions | Which area or cutoff answers the question? | Lesson 5 practice |

Finish with the integrated probability and distributions review. Separate instructor guides cover each practice and the review.

## Cases, inputs, and readings

Use this guide to connect each case to its input type and selected textbook reading.

| Case | Input used | Main 8th-edition reference |
|---|---|---|
| Restaurant selection | Existing checked 300-row CSV | Tables 2.9–2.10, pp.65–66; probability rules in §§4.1–4.4 |
| Two suppliers | Stated shares and good/bad probabilities | §4.5, pp.201–205 |
| DiCarlo Motors | Six published daily-sales frequency cells | §§5.2–5.3, pp.222–228 |
| Martin Clothing Store | Three customers, purchase probability 0.30 | §5.5, especially p.244 |
| Arrivals | Mean 10 per 15 minutes and stated process assumptions | §5.6, pp.252–255 |
| Flight duration | Uniform bounds 120–140 minutes | §6.1, pp.275–277 |
| Grear tires | Normal mean 36,500 miles and SD 5,000 miles | §6.2, especially pp.286–290 |

The Restaurant CSV retains its documented provenance and verification limits. All other inputs are explicitly identified printed aggregate values or model parameters entered in the lesson code. They are not new individual-observation datasets. Artificial simulation is labeled as such.

Read each case's English introduction followed by its equivalent Arabic narration. Statistical explanations and interpretation tasks remain in English.

## Scope and next step

The selected sequence includes five lessons, five practices, and an integrated review. It is not exhaustive coverage of Chapters 4–6: detailed permutation exercises, bivariate portfolio distributions, hypergeometric calculations, and the exponential distribution remain possible extensions.

Module 4: Sampling and estimation is the next topic; its four lessons, practice, and review are available. We will distinguish the distribution of an individual observation from the sampling distribution of a statistic.

Return to the course home.